1. $\dot{r}(t) = (-\sin t,\ a\cos t,\ b)$

$\ddot{r}(t) = (-\cos t,\ -a\sin t,\ 0)$

$\dddot{r}(t) = (\sin t,\ -a\cos t,\ 0)$

$$\kappa = \frac{|\dot{r}\times\ddot{r}|}{|\dot{r}|^3} = \frac{|(ab\sin t,\ -b\cos t,\ a)|}{(\sin^2 t + a^2\cos^2 t + b^2)^{\frac{3}{2}}} = \frac{(a^2b^2\sin^2 t + b^2\cos^2 t + a^2)^{\frac{1}{2}}}{(\sin^2 t + a^2\cos^2 t + b^2)^{\frac{3}{2}}}$$

$$\tau = \frac{\dddot{r}\cdot(\dot{r}\times\ddot{r})}{|\dot{r}\times\ddot{r}|^2} = \frac{ab}{a^2b^2\sin^2 t + b^2\cos^2 t + a^2}$$

2. $\begin{cases} x = a\cos\theta \\ y = b\sin\theta \end{cases}$

$r(\theta) = (a\cos\theta,\ b\sin\theta)$ $\quad \dot{r}(\theta) = (-a\sin\theta,\ b\cos\theta)$ $\quad \ddot{r}(\theta) = (-a\cos\theta,\ -b\sin\theta)$

$g = \dot{r}\cdot\dot{r} = a^2\sin^2\theta + b^2\cos^2\theta$

$$\frac{J(-a\sin\theta,\ b\cos\theta)^T}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}} = \frac{(-b\cos\theta,\ -a\sin\theta)^T}{(a^2\sin^2\theta + b^2\cos^2\theta)^{\frac{1}{2}}} \quad 故\ n = \frac{(-b\cos\theta,\ -a\sin\theta)}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}$$

$$h = \ddot{r}\cdot n$$

$$= (-a\cos\theta,\ -b\sin\theta)\ \frac{(-b\cos\theta,\ -a\sin\theta)}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}$$

$$= \frac{ab}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}$$

草图曲率 $h/g = \dfrac{ab}{(a^2\sin^2\theta + b^2\cos^2\theta)^{\frac{3}{2}}}$

$$(h/g)' = \frac{-ab\,\frac{3}{2}(a^2\sin^2\theta + b^2\cos^2\theta)^{\frac{1}{2}}(2a^2\sin\theta\cos\theta - 2b^2\cos\theta\sin\theta)}{(a^2\sin^2\theta + b^2\cos^2\theta)^3}$$

$$= \frac{-3ab(a^2-b^2)\sin\theta\cos\theta}{(a^2\sin^2\theta + b^2\cos^2\theta)^{\frac{5}{2}}}$$

令 $(h/g)' = 0$ 则 $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ 即 $\{(\pm a, 0), (0, \pm b)\}$ $a \neq b$. $a = b$ 时, 处处

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3. $r_\theta(\theta,\phi) = (\cos\theta\cos\phi,\ 2\cos\theta\sin\phi,\ -3\sin\theta)$

$r_\phi(\theta,\phi) = (-\sin\theta\sin\phi,\ 2\sin\theta\cos\phi,\ 0)$

$E = r_\theta \cdot r_\theta = \cos^2\theta\cos^2\phi + 4\cos^2\theta\sin^2\phi + 9\sin^2\theta = 3\cos^2\theta\sin^2\phi + 8\sin^2\theta + 1$

$F = r_\theta \cdot r_\phi = -\cos\theta\cos\phi\sin\theta\sin\phi + 4\cos\theta\sin\phi\sin\theta\cos\phi = 3\cos\theta\cos\phi\sin\theta\sin\phi$

$G = r_\phi \cdot r_\phi = \underline{\sin^2\theta\sin^2\phi} + 4\sin^2\theta\cos^2\phi = \sin^2\theta + 3\sin^2\theta\cos^2\phi$

故第一基本形式为 $I = (3\cos^2\theta\sin^2\phi + 8\sin^2\theta + 1)\,d\theta^2 + 2(3\cos\theta\cos\phi\sin\theta\sin\phi)\,d\theta\,d\phi + (\sin^2\theta + 3\sin^2\theta\cos^2\phi)\,d\phi^2$

$$\vec{N} = \frac{r_\theta \times r_\phi}{|r_\theta \times r_\phi|} = \frac{(6\sin^2\theta\cos\phi,\ 3\sin^2\theta\sin\phi,\ 2\sin\theta\cos\theta)}{(27\sin^4\theta\cos^2\phi + 9\sin^4\theta + 4\sin^2\theta\cos^2\theta)^{\frac{1}{2}}}$$

$$A \triangleq (27\sin^4\theta\cos^2\phi + 9\sin^4\theta + 4\sin^2\theta\cos^2\theta)^{\frac{1}{2}}$$

$r_{\theta\theta}(\theta,\phi) = (-\sin\theta\cos\phi,\ -2\sin\theta\sin\phi,\ -3\cos\theta)$

$r_{\theta\phi}(\theta,\phi) = (-\cos\theta\sin\phi,\ 2\cos\theta\cos\phi,\ 0)$

$r_{\phi\phi}(\theta,\phi) = (-\sin\theta\cos\phi,\ -2\sin\theta\sin\phi,\ 0)$

$$L = r_{\theta\theta} \cdot \vec{N} = -\frac{6\sin\theta}{A}$$

$$M = r_{\theta\phi} \cdot \vec{N} = 0$$

$$N = r_{\phi\phi} \cdot \vec{N} = -\frac{6\sin^3\theta}{A}$$

故第二基本形式为

$$II = -\frac{6}{(27\sin^2\theta\cos^2\phi + 5\sin^2\theta + 4)^{\frac{1}{2}}}\,d\theta^2 - \frac{6\sin^2\theta}{(27\sin^2\theta\cos^2\phi + 5\sin^2\theta + 4)^{\frac{1}{2}}}\,d\phi^2$$

4. $c=\frac{1}{100}$

$r_u(u,v)=(-\frac{1}{100}\sin u\cos v,\ -\frac{1}{100}\sin u\sin v,\ \frac{1}{100}\cos u)$

$r_v(u,v)=(-(1+\frac{1}{100}\cos u)\sin v,\ (1+\frac{1}{100}\cos u)\cos v,\ 0)$

$E=r_u\cdot r_u=\frac{1}{10000}$

$F=r_u\cdot r_v=0$

$G=r_v\cdot r_v=(1+\frac{1}{100}\cos u)^2$

故第一基本形式为 $I=\frac{1}{10000}du^2+(1+\frac{1}{100}\cos u)^2dv^2$

$\vec{N}=\frac{r_u\times r_v}{|r_u\times r_v|}=(-\cos u\cos v,-\cos u\sin v,-\sin u)$

$r_{uu}(u,v)=(-\frac{1}{100}\cos u\cos v,\ -\frac{1}{100}\cos u\sin v,\ -\frac{1}{100}\sin v)$

$r_{uv}(u,v)=(\frac{1}{100}\sin u\sin v,\ -\frac{1}{100}\sin u\cos v,\ 0)$

$r_{vv}(u,v)=(-(1+\frac{1}{100}\cos u)\cos v,\ -(1+\frac{1}{100}\cos u)\sin v,\ 0)$

$L=r_{uu}\cdot\vec{N}=\frac{1}{100}$

$M=r_{uv}\cdot\vec{N}=0$

$N=r_{vv}\cdot\vec{N}=(1+\frac{1}{100}\cos u)\cos u$

故第二基本形式为 $II=\frac{1}{100}du^2+(1+\frac{1}{100}\cos u)\cos u\,dv^2$